Numerical Solutions to Hyperbolic PDEs

Abstract

This project investigates finite difference methods for solving hyperbolic partial differential equations, focusing on the linear advection equation and the inviscid Burgers' equation. The first upwind, Lax, and Lax–Wendroff schemes are analysed theoretically, showing that all are conditionally stable for Courant numbers $C \leq 1$, with the upwind and Lax methods being first-order accurate and Lax–Wendroff second-order accurate in both space and time.

Numerical simulations confirm these theoretical results. In particular, the first upwind and Lax–Wendroff schemes became unstable when $\Delta x = 5$ and $\Delta t = 0.021$, as expected since the Courant number exceeded the stability limit. For the inviscid Burgers' equation, the Lax and MacCormack methods were implemented with modified flux terms to account for nonlinearity. Solutions with $\Delta t = 0.14$ became unstable over time, whereas $\Delta t = 0.1$ and 0.05 remained stable, with $\Delta t = 0.1$ providing the best balance between stability and accuracy. Comparison with exact solutions at $t = 2.4$ showed that MacCormack and Lax–Wendroff methods better captured steep gradients, while the Lax method introduced noticeable numerical diffusion.

Overall, the study highlights the trade-offs between stability, accuracy, and computational efficiency, and demonstrates the importance of selecting appropriate discretisation parameters when applying finite difference schemes to linear and nonlinear hyperbolic PDEs.

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1 Introduction

Partial differential equations (PDEs) play a fundamental role in modelling a wide range of physical phenomena, including fluid flow, wave propagation, and transport processes. In many cases, analytical solutions are difficult or impossible to obtain, making numerical methods essential tools for approximating solutions. This project focuses on the numerical solution of hyperbolic PDEs, specifically the linear advection equation and the inviscid Burgers' equation.

We begin by considering several finite difference schemes commonly used for hyperbolic problems, including the first upwind method, the Lax method, and the Lax–Wendroff method. These schemes are analysed theoretically using Von Neumann stability analysis to determine conditions under which numerical solutions remain stable. In addition, consistency analysis is performed to verify that each scheme correctly approximates the underlying differential equation as the grid spacing tends to zero. The relationship between the MacCormack method and the Lax–Wendroff scheme is also examined, highlighting their equivalence under certain conditions.

A key parameter in the analysis of numerical methods for hyperbolic equations is the Courant number, which governs the relationship between temporal and spatial discretisation. Its influence on stability and accuracy is investigated throughout this project.

Following the theoretical analysis, the numerical implementation of these methods is carried out. The first upwind, Lax, Lax–Wendroff, and BTCS schemes are applied to the linear advection equation to study their stability and accuracy properties. The Lax and MacCormack methods are then extended to the nonlinear inviscid Burgers' equation, requiring appropriate modifications to account for the nonlinearity of the flux term.

The performance of the numerical schemes is evaluated by examining how solutions evolve over time for different time step sizes, as well as by comparing numerical results with exact solutions at specific times. Particular attention is given to stability, numerical diffusion, and the presence of non-physical oscillations. This allows for a comprehensive comparison of the methods and provides insight into their suitability for solving hyperbolic PDEs.

1.1 First Upwind differencing method Von-Neumann Stability Analysis

We start with the First Upwind differencing method

$$U_k^{n+1} = U_k^n - c\left(U_k^n - U_{k-1}^n\right) \qquad \text{where} \qquad c := \frac{a\Delta t}{\Delta x}$$

Assume a Fourier mode solution and substitute $U_k^n \to \xi^n e^{i\theta k}$. Then

$$\xi^{n+1} e^{i\theta k} = \xi^n e^{i\theta k} - c\left(\xi^n e^{i\theta k} - \xi^n e^{i\theta(k-1)}\right).$$

Divide by $\xi^n e^{i\theta k}$ and use Euler's formula

$$\xi = 1 - c + c\cos\theta - ic\sin\theta$$

Taking modulus

$$\begin{aligned} |\xi|^2 &= (1 - c + c\cos\theta)^2 + c^2\sin^2\theta \\ &= 1 - 2c + 2c^2 + 2c(1-c)\cos\theta. \end{aligned}$$

We define

$$f(\theta) = 1 - 2c + 2c^2 + 2c(1-c)\cos\theta$$

To find the extrema of $f(\theta)$, we differentiate:

$$f'(\theta) = 2c(c-1)\sin\theta = 0.$$

Hence,

$$\sin\theta = 0 \;\Rightarrow\; \theta = 0 + n\pi, \quad n \in \mathbb{Z}.$$

At these points $\cos\theta = \pm 1$. Thus it suffices to check these cases.
If $\cos\theta = 1$, then

$$|\xi| = 1.$$

If $\cos\theta = -1$, then

$$|\xi| = \sqrt{1 - 4c + 4c^2} = |1 - 2c|.$$

Now,

$$|1-2c| \le 1 \iff -1 \le 1 - 2c \le 1 \iff 0 \le c \le 1.$$

Since $c = \frac{a\Delta t}{\Delta x}$ with $a > 0$, we have $c \ge 0$. Hence the stability condition reduces to

$$c \le 1.$$

Therefore, the scheme is **conditionally stable**.

1.2 Lax method Von-Neumann Stability Analysis

We start with the Lax method method

$$U_k^{n+1} = \frac{1}{2}\left(U_{k+1}^n + U_{k-1}^n\right) - \frac{c}{2}\left(U_{k+1}^n - U_{k-1}^n\right) \qquad \text{where} \qquad c := \frac{a\Delta t}{\Delta x}$$

Assume a Fourier mode solution and substitute $U_k^n \to \xi^n$

$$\xi^{n+1}e^{i\theta k} = \frac{1}{2}\left(\xi^n e^{i\theta(k+1)} + \xi^n e^{i\theta(k-1)}\right) - \frac{c}{2}\left(\xi^n e^{i\theta(k+1)} - \xi^n e^{i\theta(k-1)}\right).$$

Divide by $\xi^n e^{i\theta k}$

$$\xi = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) - \frac{c}{2}\left(e^{i\theta} - e^{-i\theta}\right).$$

and use substitutions $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ and $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$

$$\xi = \cos\theta - ic\sin\theta.$$

We take modulus and use $\sin^2\theta = 1 - \cos^2\theta$

$$|\xi|^2 = \cos^2\theta + c^2\sin^2\theta = c^2 + (1-c^2)\cos^2\theta.$$

Since $|\xi|^2 = c^2 + (1-c^2)\cos^2\theta$ is linear in $\cos^2\theta \in [0,1]$, the maximum occurs at $\cos^2\theta = 0$ or 1.
If $\cos\theta = 1$, then

$$|\xi| = 1.$$

If $\cos\theta = 0$, then

$$|\xi| = |c|$$

Since $c = \frac{a\Delta t}{\Delta x}$ with $a > 0$, we have $c \ge 0$. Hence the stability condition reduces to

$$c \le 1.$$

Therefore, the scheme is **conditionally stable**.

1.3 Equivalence of the MacCormack and Lax–Wendroff Methods

We want to show the MacCormack method consisting of
Predictor step:

$$U_k^* = U_k^n - c\left(U_{k+1}^n - U_k^n\right)$$

Corrector step:

$$U_k^{n+1} = \frac{1}{2}\left[(U_k^n + U_k^*) - c\left(U_k^* - U_{k-1}^*\right)\right]$$

is equivalent to the Lax-Wendroff method

$$U_k^{n+1} = U_k^n - \frac{c}{2}\left(U_{k+1}^n - U_{k-1}^n\right) + \frac{c^2}{2}\left(U_{k-1}^n - 2U_k^n + U_{k+1}^n\right) \qquad \text{where} \qquad c := \frac{a\Delta t}{\Delta x}.$$

We start by substituting the predictor step into the corrector step.

$$U_k^{n+1} = \frac{1}{2}\left[\left(U_k^n + U_k^n - c(U_{k+1}^n - U_k^n)\right) - c\left((U_k^n - c(U_{k+1}^n - U_k^n)) - (U_{k-1}^n - c(U_k^n - U_{k-1}^n))\right)\right].$$

Now group terms:

$$U_k^{n+1} = \frac{1}{2}\Big[2U_k^n - c\left(U_{k+1}^n - U_{k-1}^n\right) + c^2\left(U_{k+1}^n - 2U_k^n + U_{k-1}^n\right)\Big].$$

which is exactly Lax-Wendroff method once $\frac{1}{2}$ is distributed

$$U_k^{n+1} = U_k^n - \frac{c}{2}\left(U_{k+1}^n - U_{k-1}^n\right) + \frac{c^2}{2}\left(U_{k-1}^n - 2U_k^n + U_{k+1}^n\right)$$

1.4 Lax-Wendroff method Von-Neumann Stability Analysis

We start with the Lax-Wendroff method

$$U_k^{n+1} = U_k^n - \frac{c}{2}\left(U_{k+1}^n - U_{k-1}^n\right) + \frac{c^2}{2}\left(U_{k-1}^n - 2U_k^n + U_{k+1}^n\right) \qquad \text{where} \qquad c := \frac{a\Delta t}{\Delta x}.$$

Assume a Fourier mode solution and substitute $U_k^n \to \xi^n e^{i\theta k}$. Then

$$\xi^{n+1}e^{i\theta k} = \xi^n e^{i\theta k} - \frac{c}{2}\xi^n e^{i\theta k}\left(e^{i\theta} - e^{-i\theta}\right) + \frac{c^2}{2}\xi^n e^{i\theta k}\left(e^{-i\theta} - 2 + e^{i\theta}\right).$$

Divide by $\xi^n e^{i\theta k}$ and use substitutions $e^{i\theta} + e^{-i\theta} = 2\cos\theta$ and $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$

$$\xi = 1 - ic\sin\theta + c^2(\cos\theta - 1).$$

Take modulus

$$|\xi|^2 = \left(1 + c^2(\cos\theta - 1)\right)^2 + (c\sin\theta)^2.$$

Now we expand and group

$$|\xi|^2 = 1 - 2c^2 + c^4 + (2c^2 - 2c^4)\cos\theta + c^4\cos^2\theta + c^2\sin^2\theta$$

and define a function

$$f(\theta) := |\xi|^2 = 1 - 2c^2 + c^4 + (2c^2 - 2c^4)\cos\theta + c^4\cos^2\theta + c^2\sin^2\theta.$$

Differentiate $f(\theta)$ with respect to θ

$$f'(\theta) = -(2c^2 - 2c^4)\sin\theta - 2c^4\cos\theta\sin\theta + 2c^2\sin\theta\cos\theta$$

and simplify

$$f'(\theta) = 2c^2\sin\theta\left[-1 + c^2 + (1-c^2)\cos\theta\right] = 0.$$

Then

$$\sin\theta = 0 \implies \theta = n\pi, \quad n \in \mathbb{Z},$$

or

$$\cos\theta = 1 \implies \theta = 2n\pi, \quad n \in \mathbb{Z}.$$

We check the values at the critical points:

$$f(0) = 1 - 2c^2 + c^4 + (2c^2 - 2c^4)(1) + c^4(1) + c^2(0) = 1.$$

and

$$f(\pi) = 1 - 2c^2 + c^4 + (2c^2 - 2c^4)(-1) + c^4(1) + c^2(0) = 1 - 4c^2 + 4c^4 = (1-2c^2)^2.$$

Now if

$$\begin{aligned}(1-2c^2)^2 \le 1 &\iff |1-2c^2| \le 1\\ &\iff -1 \le 1 - 2c^2 \le 1\\ &\iff -2 \le -2c^2 \le 0\\ &\iff 0 \le c^2 \le 1\\ &\iff |c| \le 1.\end{aligned}$$

Therefore, the scheme is **conditionally stable**.

1.5 First Upwind differencing method Consistency Analysis

We start with the first-order upwind differencing method

$$\frac{U_k^{n+1} - U_k^n}{\Delta t} = -a\frac{U_k^n - U_{k-1}^n}{\Delta x}.$$

We approximate each finite difference using Taylor expansions:

$$\frac{u(x_k, t_{n+1}) - u(x_k, t_n)}{\Delta t} = \frac{\partial u}{\partial t}(x_k, t_n) + O(\Delta t), \tag{1}$$

$$\frac{u(x_k, t_n) - u(x_{k-1}, t_n)}{\Delta x} = \frac{\partial u}{\partial x}(x_k, t_n) + O(\Delta x). \tag{2}$$

Substituting (3) and (2) into the scheme gives

$$\frac{\partial u}{\partial t}(x_k, t_n) + O(\Delta t) = -a\left(\frac{\partial u}{\partial x}(x_k, t_n) + O(\Delta x)\right).$$

Rearranging,

$$\frac{\partial u}{\partial t}(x_k, t_n) + a\frac{\partial u}{\partial x}(x_k, t_n) + O(\Delta t, \Delta x) = 0.$$

Using the PDE

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = 0,$$

we conclude that the truncation error is

$$O(\Delta t, \Delta x).$$

Hence, the first-order upwind differencing method is **first-order accurate in both space and time**.

1.6 Lax method Consistency Analysis

We start with the Lax scheme

$$U_k^{n+1} = \frac{1}{2}\left(U_{k+1}^n + U_{k-1}^n\right) - \frac{c}{2}\left(U_{k+1}^n - U_{k-1}^n\right), \qquad \text{where} \qquad c := \frac{a\Delta t}{\Delta x}.$$

Subtracting U_k^n and dividing by Δt, we obtain

$$\frac{U_k^{n+1} - U_k^n}{\Delta t} = -a\frac{U_{k+1}^n - U_{k-1}^n}{2\Delta x} + \frac{\left(U_{k+1}^n - 2U_k^n + U_{k-1}^n\right)}{2\Delta t}.$$

We approximate each finite difference using Taylor expansions

$$\frac{U_k^{n+1} - U_k^n}{\Delta t} = \frac{\partial u}{\partial t}(x_k, t_n) + O(\Delta t), \tag{3}$$

$$\frac{U_{k+1}^n - U_{k-1}^n}{2\Delta x} = \frac{\partial u}{\partial x}(x_k, t_n) + O(\Delta x^2), \tag{4}$$

$$U_{k+1}^n - 2U_k^n + U_{k-1}^n = \Delta x^2 \frac{\partial^2 u}{\partial x^2}(x_k, t_n) + O(\Delta x^4). \tag{5}$$

Now,

$$\text{TE} = \frac{U_k^{n+1} - U_k^n}{\Delta t} + a\frac{U_{k+1}^n - U_{k-1}^n}{2\Delta x} - \frac{\left(U_{k+1}^n - 2U_k^n + U_{k-1}^n\right)}{2\Delta t} \tag{6}$$

Substituting (3)–(5) into (6), we obtain

$$= \frac{\partial u}{\partial t}(x_k, t_n) + O(\Delta t) + a\left(\frac{\partial u}{\partial x}(x_k, t_n) + O(\Delta x^2)\right) - \frac{\Delta x^2}{2\Delta t}\frac{\partial^2 u}{\partial x^2}(x_k, t_n) + O\left(\frac{\Delta x^4}{\Delta t}\right).$$

Using the PDE $u_t + au_x = 0$ and grouping dominant errors, we obtain

$$\text{TE} = -\frac{\Delta x^2}{2\Delta t}\frac{\partial^2 u}{\partial x^2}(x_k, t_n) + O(\Delta t, \Delta x^2).$$

Now,

$$\left|\frac{\Delta x^2}{2\Delta t}\frac{\partial^2 u}{\partial x^2}(x_k, t_n)\right| \leq C\frac{\Delta x^2}{\Delta t} = O\left(\frac{\Delta x^2}{\Delta t}\right).$$

If $\Delta t \sim \Delta x$, then

$$\frac{\Delta x^2}{\Delta t} = O(\Delta x).$$

we conclude that the truncation error is

$$O(\Delta t, \Delta x).$$

Hence, the Lax method is **first-order accurate in both space and time**.

1.7 Lax-Wendroff method Consistency Analysis

We start with the Lax-Wendroff method

$$\frac{U_k^{n+1} - U_k^n}{\Delta t} = -a\left[\frac{U_{k+1}^n - U_{k-1}^n}{2\Delta x}\right] + \frac{1}{2}a^2\Delta t\left[\frac{U_{k-1}^n - 2U_k^n + U_{k+1}^n}{(\Delta x)^2}\right]$$

We approximate each finite difference using Taylor expansions:

$$\frac{U_k^{n+1} - U_k^n}{\Delta t} = \frac{\partial u}{\partial t}(x_k, t_n) + \frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2}(x_k, t_n) + O(\Delta t^2), \tag{7}$$

$$\frac{U_{k+1}^n - U_{k-1}^n}{2\Delta x} = \frac{\partial u}{\partial x}(x_k, t_n) + O(\Delta x^2), \tag{8}$$

$$\frac{U_{k+1}^n - 2U_k^n + U_{k-1}^n}{(\Delta x)^2} = \frac{\partial^2 u}{\partial x^2}(x_k, t_n) + O(\Delta x^2). \tag{9}$$

Now,

$$\mathrm{TE} = \frac{U_k^{n+1} - U_k^n}{\Delta t} + a\left[\frac{U_{k+1}^n - U_{k-1}^n}{2\Delta x}\right] - \frac{1}{2}a^2\Delta t\left[\frac{U_{k-1}^n - 2U_k^n + U_{k+1}^n}{(\Delta x)^2}\right] \tag{10}$$

Substituting (7)–(9) into (10), we obtain

$$= \frac{\partial u}{\partial t}(x_k, t_n) + \frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2}(x_k, t_n) + O(\Delta t^2) + a\Big(\frac{\partial u}{\partial x}(x_k, t_n) + O(\Delta x^2)\Big) - \frac{1}{2}a^2\Delta t\Big(\frac{\partial^2 u}{\partial x^2}(x_k, t_n) + O(\Delta x^2)\Big)$$

Using the PDE $u_t + au_x = 0$ and grouping dominant errors, we obtain

$$\mathrm{TE} = O(\Delta t^2, \Delta x^2) + \frac{\Delta t}{2}\left(\frac{\partial^2 u}{\partial t^2}(x_k, t_n) - a^2\frac{\partial^2 u}{\partial x^2}(x_k, t_n)\right) \; + \; O(\Delta t \Delta x^2)$$

Differentiating our PDE again we obtain $\frac{\partial^2 u}{\partial t^2} = a^2\frac{\partial^2 u}{\partial x^2}$.
Hence

$$\mathrm{TE} = O(\Delta t^2, \Delta x^2) \; + \; O(\Delta t \Delta x^2)$$

Since TE is dominated by $O(\Delta t^2, \Delta x^2)$ for any choice of Δt, as $\Delta t\, \Delta x^2$ is always higher-order, we conclude

$$\mathrm{TE} = O(\Delta t^2, \Delta x^2).$$

That is, the Lax-Wendroff method is **second-order accurate in both space and time**

2 Advection equation (Linear problem)

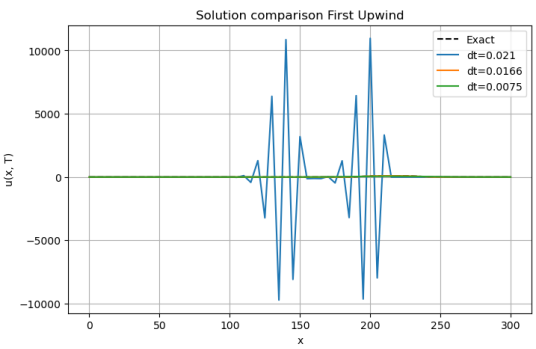
2.1 Investigating the effect of the Courant number

In Figures 1–3, we investigate the effect of the Courant number on the numerical solution of the advection equation at $t \approx 0.45$. This is done by fixing $\Delta x = 5$ and varying Δt, which changes the Courant number $c = \frac{a\Delta t}{\Delta x}$.

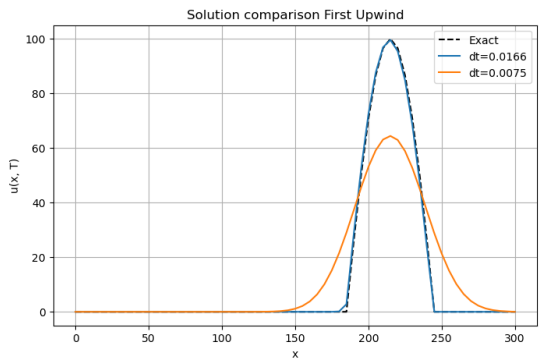
For the **First Upwind** differencing method (Figure 1), we observe that the scheme is unstable for $\Delta t = 0.021$, corresponding to $c = 1.26$. This violates the stability condition $c \leq 1$, explaining the observed numerical instability.

For smaller time steps, $\Delta t = 0.0166$ and $\Delta t = 0.0075$, the method is stable. Comparing with the exact solution, both cases show stable behaviour, with $\Delta t = 0.0166$ giving a more accurate approximation. The solution for $\Delta t = 0.0075$ appears more diffused, indicating increased numerical dissipation.
For the **Lax–Wendroff** method (Figure 2), we again observe instability when $\Delta t = 0.021$, corresponding to $c = 1.26$. This is expected, as the method is conditionally stable for $c \leq 1$.

For $\Delta t = 0.0166$ and 0.0075, the method is stable. Both agree reasonably with the exact solution, though $\Delta t = 0.0166$ is more accurate, while $\Delta t = 0.0075$ shows greater diffusion.

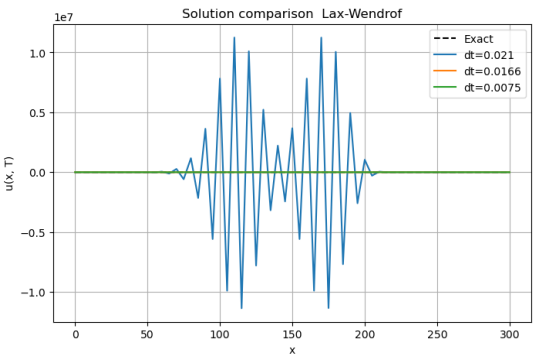


((a)) $\Delta x = 5$ and $\Delta t \in \{0.021, 0.0166, 0.0075\}$

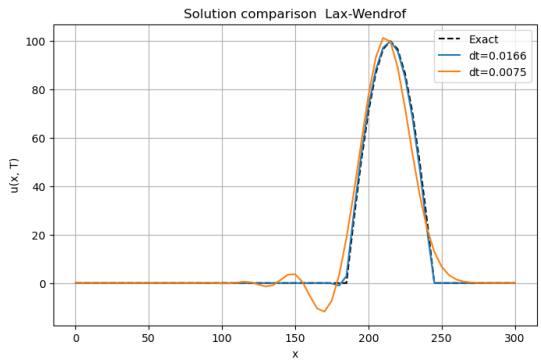


((b)) $\Delta x = 5$ and $\Delta t \in \{0.0166, 0.0075\}$

Figure 1: First Upwind differencing method numerical solutions of the advection equation at $t \approx 0.45$ for different Courant numbers, with $\Delta x = 5$ and $\Delta t \in \{0.021, 0.0166, 0.0075\}$, compared to the exact solution.



((a)) $\Delta x = 5$ and $\Delta t \in \{0.021, 0.0166, 0.0075\}$



((b)) $\Delta x = 5$ and $\Delta t \in \{0.0166, 0.0075\}$

Figure 2: Lax-Wendroff method numerical solutions of the advection equation at $t \approx 0.45$ for different Courant numbers, with $\Delta x = 5$ and $\Delta t \in \{0.021, 0.0166, 0.0075\}$, compared to the exact solution.

For **BTCS** we have stability for all $\Delta t \in 0.021, 0.0166, 0.0075$ (Figure 3), which is expected as BTCS is unconditionally stable, and the solution improves as Δt decreases, indicating increased accuracy.

For the BTCS method, the accuracy can be improved by refining the spatial discretisation. The scheme is first-order accurate in time and second-order accurate in space, i.e. $\mathcal{O}(\Delta t, (\Delta x)^2)$.

Even though the spatial discretisation is of higher order, the temporal error was initially dominant for the chosen parameters. This explains why reducing Δt led to a more noticeable improvement in the solution. However, decreasing Δt alone can introduce non-physical oscillations, which are reduced when Δx is also decreased, as illustrated in Figure 4.

Therefore, while refining Δt improves accuracy, reducing Δx is necessary to obtain a smoother and more physically accurate solution, and the best results are achieved when both are refined appropriately.

b) The exact solution $u(x,t) = u_0(x - at)$ cannot be used for all $t > 0$ on a finite domain, as it does not, in general, satisfy the imposed boundary conditions once the advected profile reaches the domain boundaries. It is therefore only valid up to the time at which boundary effects become significant, as

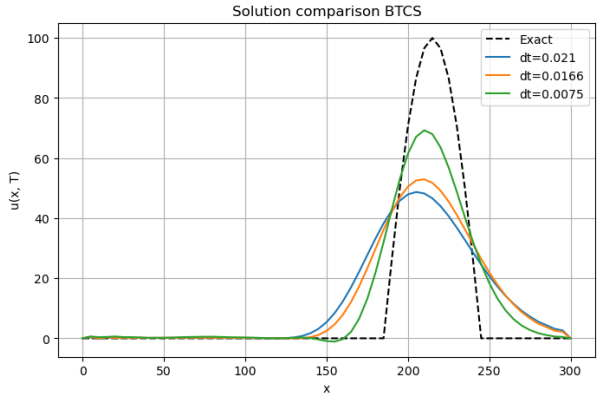
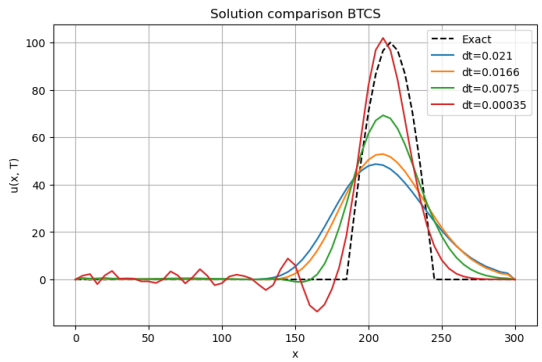
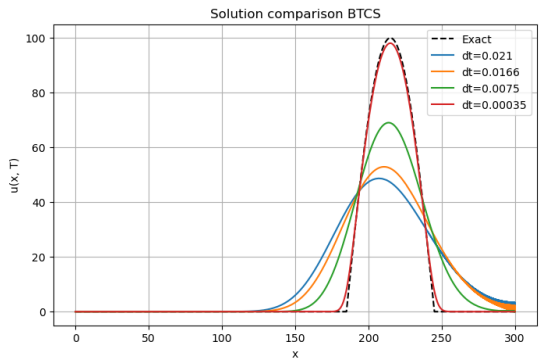


Figure 3: BTCS method numerical solutions of the advection equation at $t \approx 0.45$ for different Courant numbers, with $\Delta x = 5$ and $\Delta t \in \{0.021, 0.0166, 0.0075\}$, compared to the exact solution.



((a)) $\Delta x = 5$ and $\Delta t \in \{0.021, 0.0166, 0.0075, 0.00035\}$



((b)) $\Delta x = 0.05$, $\Delta t \in \{0.021, 0.0166, 0.0075, 0.00035\}$

Figure 4: BTCS numerical solutions showing the effect of spatial refinement.

illustrated in Figure 5.

3 Inviscid Burgers' equation (Non linear problem)

We aim to modify the Lax and MacCormack methods, originally developed for the advection equation, so that they can be applied to the inviscid Burgers' equation.

3.1 Lax method for the the inviscid Burgers' equation

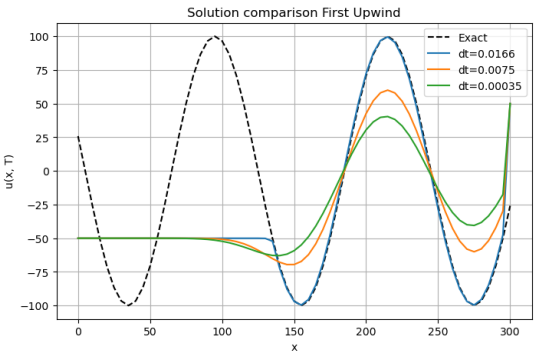
We start by comparing advection and inviscid Burgers' equations

$$\text{Advection:} \quad \frac{\partial u}{\partial t} + \frac{\partial (au)}{\partial x} = 0$$

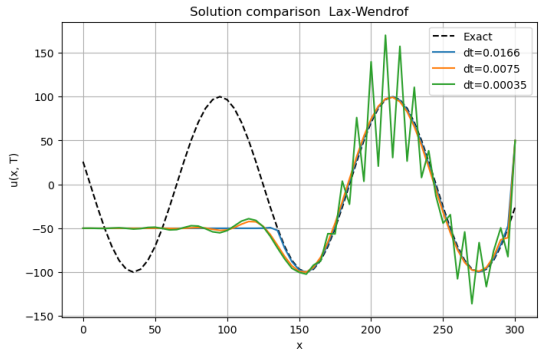
$$\text{Inviscid Burgers':} \quad \frac{\partial u}{\partial t} + \frac{\partial}{\partial x}\left(\frac{u^2}{2}\right) = 0$$

If we write both equations in conservation form

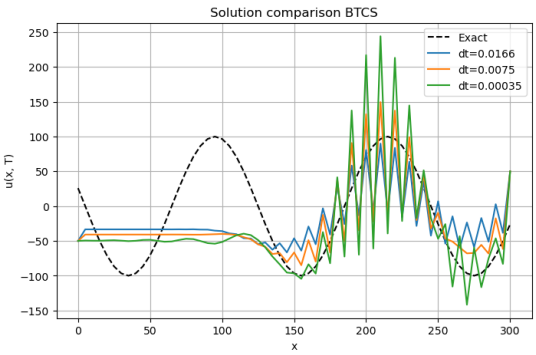
$$u_t + f(u)_x = 0,$$



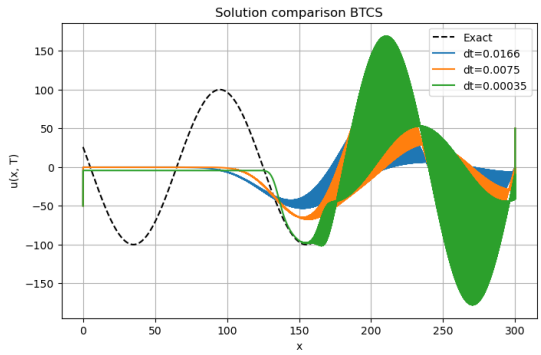
First Upwind differencing method: $\Delta x = 5$, $\Delta t \in \{0.0166, 0.0075, 0.00035\}$



((a)) Lax–Wendroff method: $\Delta x = 5$, $\Delta t \in \{0.0166, 0.0075, 0.00035\}$



((b)) BTCS method: $\Delta x = 5$, $\Delta t \in \{0.0166, 0.0075, 0.00035\}$



((c)) BTCS method: $\Delta x = 0.05$, $\Delta t \in \{0.0166, 0.0075, 0.00035\}$

Figure 5: Comparison of methods with the exact solution $u(x,t) = u_0(x - at)$ for $0 < t < 0.45$.

then for advection $f(u) = au$, while for Burgers' equation $f(u) = \frac{u^2}{2}$. Then

Lax for Advection: $$U_k^{n+1} = \frac{1}{2}\left(U_{k+1}^n + U_{k-1}^n\right) - \frac{\Delta t}{2\Delta x}\left[f(U_{k+1}^n) - f(U_{k-1}^n)\right], \quad \text{where } f(u) = au.$$

Replacing $f(u) = au$ with $f(u) = \frac{u^2}{2}$, we obtain:

Lax for Inviscid Burgers': $$U_k^{n+1} = \frac{1}{2}\left(U_{k+1}^n + U_{k-1}^n\right) - \frac{\Delta t}{2\Delta x}\left(f(U_{k+1}^n) - f(U_{k-1}^n)\right) \quad \text{where } f(u) = \frac{(u)^2}{2}.$$

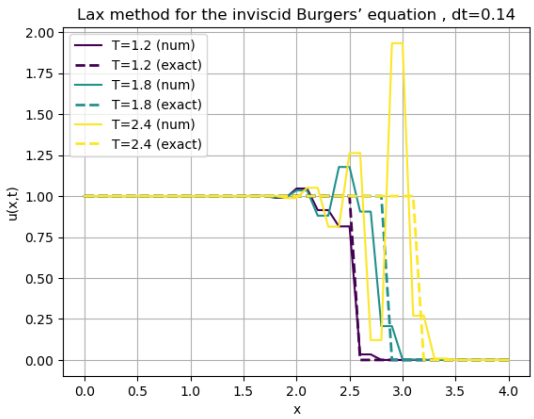
Further,

$$U_k^{n+1} = \frac{1}{2}\left(U_{k+1}^n + U_{k-1}^n\right) - \frac{\Delta t}{\Delta x}\left(\frac{(U_{k+1}^n)^2}{2} - \frac{(U_{k-1}^n)^2}{2}\right).$$

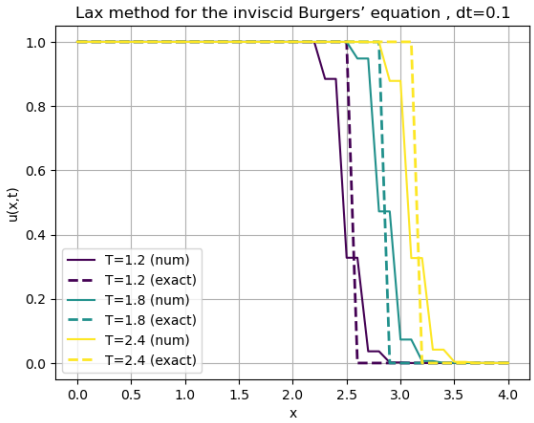
Finally,

$$U_k^{n+1} = \frac{1}{2}\left(U_{k+1}^n + U_{k-1}^n\right) - \frac{\Delta t}{4\Delta x}\left((U_{k+1}^n)^2 - (U_{k-1}^n)^2\right).$$

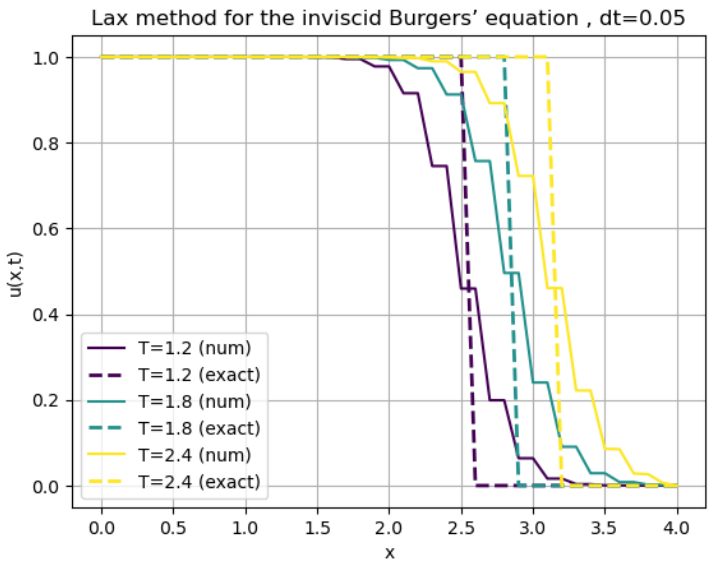
In Figure 6, we present Lax method solutions of the inviscid Burgers' equation with $\Delta x = 0.1$ for time steps $\Delta t \in 0.14, 0.1, 0.05$ at $t = 1.2$, 1.8, and 2.4. The method appears unstable for $\Delta t = 0.14$, with errors growing as time increases. In contrast, for $\Delta t = 0.1$ and 0.05, the method remains stable, and the solutions at $t = 1.2$ and $t = 1.8$ show similar behaviour. Overall, $\Delta t = 0.1$ provides the most consistent



((a)) $\Delta x = 0.1,\ \Delta t = 0.14$



((b)) $\Delta x = 0.1,\ \Delta t = 0.1$



((c)) $\Delta x = 0.1,\ \Delta t = 0.05$

Figure 6: Lax method solutions of the inviscid Burgers' equation with $\Delta x = 0.1$ for $\Delta t \in \{0.14, 0.1, 0.05\}$ at times $t = 1.2$, 1.8, and 2.4.

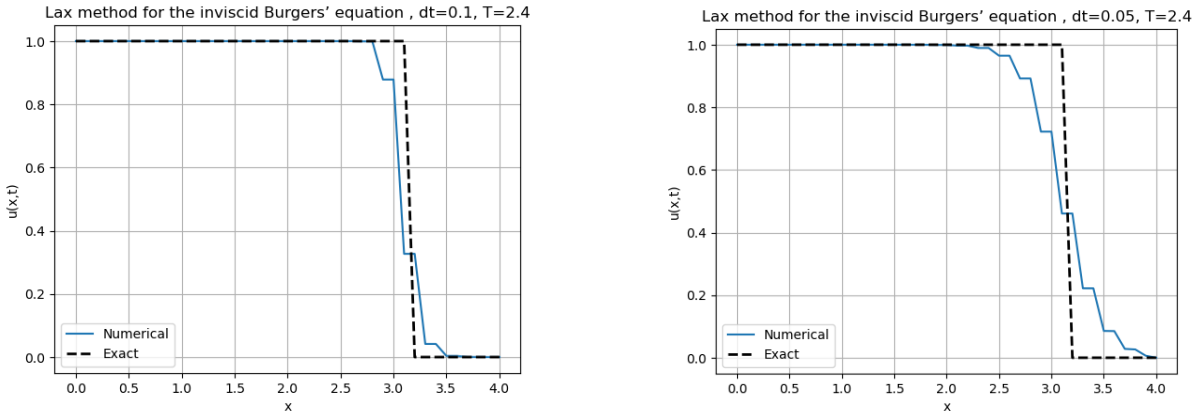
and well-resolved solution across the times considered.

In Figure 7, we plot Lax method solutions of the inviscid Burgers' equation with $\Delta x = 0.1$ at $t = 2.4$, alongside the exact solution. We observe that $\Delta t = 0.1$ provides a closer fit to the exact solution than $\Delta t = 0.05$.

3.2 MacCormack method for the the inviscid Burgers' equation

To extend the MacCormack method from the advection equation to the inviscid Burgers' equation, we first rewrite the scheme in conservation form using the flux $f(u) = au$.

$$\text{Predictor step:} \quad U_k^* = U_k^n - c\left(U_{k+1}^n - U_k^n\right) \implies U_k^* = U_k^n - \frac{\Delta t}{\Delta x}\left[f(U_{k+1}^n) - f(U_k^n)\right]$$



((a)) $\Delta t = 0.1$ ((b)) $\Delta t = 0.05$

Figure 7: Lax method solutions of the inviscid Burgers' equation with $\Delta x = 0.1$ at $t = 2.4$, compared with the exact solution.

$$\text{Corrector step:} \quad U_k^{n+1} = \frac{1}{2}\left[U_k^n + U_k^* - c\left(U_k^* - U_{k-1}^*\right)\right] \implies U_k^{n+1} = \frac{1}{2}\left[U_k^n + U_k^* - \frac{\Delta t}{\Delta x}\left(f(U_k^*) - f(U_{k-1}^*)\right)\right].$$

Now, replace this flux with $f(u) = \frac{u^2}{2}$ for Burgers' equation to obtain

$$\text{Predictor step:} \quad U_k^* = U_k^n - \frac{\Delta t}{2\Delta x}\left[(U_{k+1}^n)^2 - (U_k^n)^2\right].$$

$$\text{Corrector step:} \quad U_k^{n+1} = \frac{1}{2}\left[U_k^n + U_k^* - \frac{\Delta t}{2\Delta x}\left((U_k^*)^2 - (U_{k-1}^*)^2\right)\right].$$

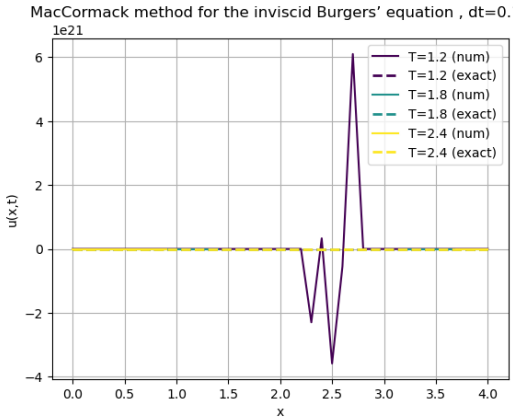
In Figure 8, we present MacCormack method solutions of the inviscid Burgers' equation with $\Delta x = 0.1$ for time steps $\Delta t \in 0.14, 0.1, 0.05$ at $t = 1.2$, 1.8, and 2.4. The method appears unstable for $\Delta t = 0.14$, particularly at $t = 1.2$, where significant oscillations are observed, with smaller oscillations persisting at later times. Similar oscillatory behaviour is also present for $\Delta t = 0.1$ and 0.05, although it is less pronounced. For $\Delta t = 0.05$, slight instability is visible near the inflection point of the solution. Overall, $\Delta t = 0.1$ provides the most accurate and well-behaved approximation.

In Figure 9 we plot MacCormack method solutions of the inviscid Burgers' equation with $\Delta x = 0.1$ at $t = 2.4$, alongside the exact solution. We observe that $\Delta t = 0.1$ provides a closer fit to the exact solution than $\Delta t = 0.05$.

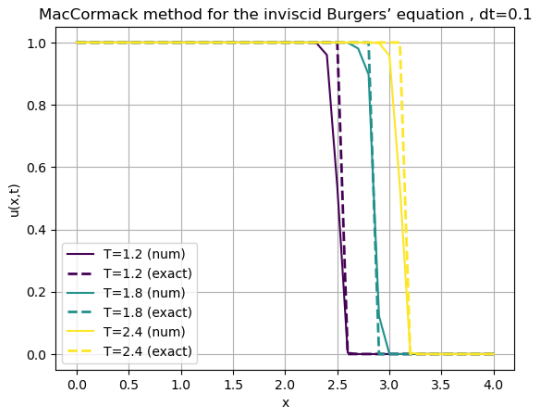
4 Conclusion

In this project, several finite difference schemes were analysed and implemented for linear and nonlinear hyperbolic partial differential equations, including the linear advection equation and the inviscid Burgers' equation. The first upwind, Lax, and Lax–Wendroff methods were studied theoretically through Von Neumann stability and consistency analyses, showing that all are conditionally stable for Courant numbers $C \leq 1$, with the upwind and Lax methods being first-order accurate and Lax–Wendroff second-order accurate in both space and time.

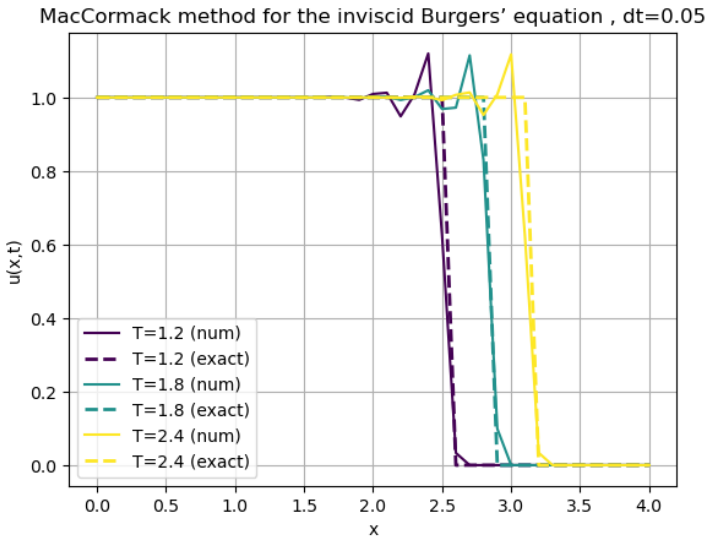
Numerical experiments confirmed these theoretical predictions. The first upwind and Lax–Wendroff schemes became unstable when Δx and Δt produced a Courant number exceeding 1, demonstrating the importance of satisfying the CFL condition. For the inviscid Burgers' equation, modifications to the



((a)) $\Delta x = 0.1,\ \Delta t = 0.14$



((b)) $\Delta x = 0.1,\ \Delta t = 0.1$

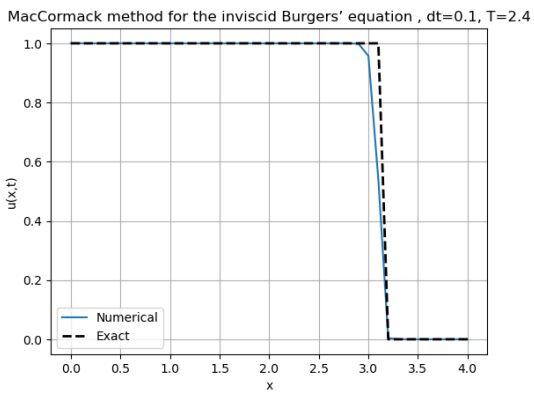


((c)) $\Delta x = 0.1,\ \Delta t = 0.05$

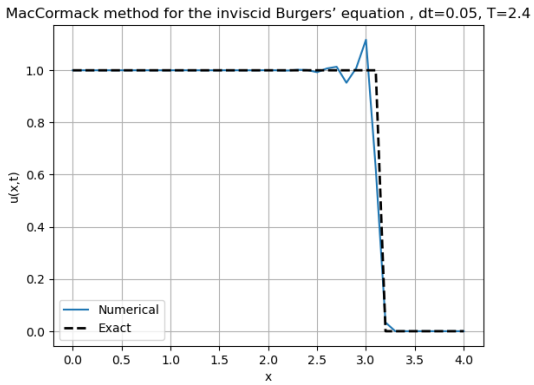
Figure 8: MacCormack method solutions of the inviscid Burgers' equation with $\Delta x = 0.1$ for $\Delta t \in \{0.14, 0.1, 0.05\}$ at times $t = 1.2$, 1.8, and 2.4.

Lax and MacCormack methods allowed them to handle nonlinearity. Solutions revealed that $\Delta t = 0.1$ provided the best balance between stability and accuracy, with MacCormack and Lax–Wendroff methods capturing steep gradients effectively, while the Lax method exhibited noticeable numerical diffusion.

Overall, the study highlights the trade-offs between stability, accuracy, and computational efficiency, and underscores the need to carefully choose discretisation parameters when applying finite difference methods to hyperbolic PDEs.



((a)) $\Delta t = 0.1$



((b)) $\Delta t = 0.05$

Figure 9: MacCormack method solutions of the inviscid Burgers' equation with $\Delta x = 0.1$ at $t = 2.4$, compared with the exact solution.